

Probability and Statistics for Engineers (Stat 2171)

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Course Outline

1. Basic Concepts, methods of data collection and presentation

- Introduction
 - Definition and classification of Statistics
 - Stages in statistical investigation
 - Definition of Some Basic terms
 - Applications, uses and limitations of statistics
 - Types of variables and measurement scales
- Methods of data collection and presentation
 - Methods of data collection
 - Sources and types of data
 - Methods of data presentation
 - Frequency distributions
 - Diagrammatic and/or graphical presentation of data

2. Summarizing of Data

- Measures of central tendency
- Types of measures of central tendency
 - mean, mode, median
- Measures of location: quantiles
- Measures of dispersion/variation
 - range, variance, standard deviation and coefficient of variation
- Standard scores

3. Elementary Probability

- Deterministic and non-deterministic models
- Review of set theory: sets, union, intersection, complementation, De Morgan's rules
- Random experiments, sample space and events
- Finite sample spaces and equally likely outcomes
- Counting techniques
- Definitions of probability
- Derived theorems of probability

4. **Conditional Probability and Independence**

- Conditional probability
- Multiplication theorem, Bayes' Theorem, total probability theorem
- Independent events

5. **One-dimensional Random Variables**

- Random variable: definition and distribution function
- Discrete random variables
- Continuous random variables
- Cumulative distribution function and its properties

6. **Functions of Random Variables**

- Equivalent events
- Functions of discrete random variables and their distributions
- Functions of continuous random variables and their distributions

7. Two dimensional Random Variables

- Two dimensional random variables
- Joint distributions for discrete and continuous random variables
- Marginal and conditional distributions
- Independent random variables
- Distributions of functions of two dimensional random variables

8. Expectation

- Expectation of a random variable
- Expectation of a function of a random variable
- Properties of expectation
- Variance of a random variable and its properties
- Moments and moment generating function
- Chebychev's Inequality
- Covariance, correlation Coefficient

9. Common Probability distributions

- Common Discrete Distributions and their Properties
 - Binomial distribution
 - Poisson distribution
 - Geometric distribution
- Common Continuous Distributions and their Properties
 - Uniform distribution
 - Normal distribution
 - Exponential distribution

10. Simple Linear Regression and Correlation

- Introduction
- Fitting simple linear regression
- Covariance and the correlation coefficient
- Rank correlation coefficient

1.1 Introduction

Definition of Statistics

- Plural form
 - numerical facts and figures collected for a certain purposes
 - aggregates of numerical expressed facts (figures) collected in a systematic manner for a predetermined purpose
- Singular form
 - systematic collection and interpretation of numerical data to make a decision
 - the science of collecting, organizing, presenting, analyzing and interpreting numerical data to make decision on the bases of such analysis

Classification of Statistics

- **Descriptive Statistics**

- Mainly concerned with the methods and techniques used in collection, organization, presentation, and analysis of a set of data without making any conclusions or inferences.
 - Gathering data
 - Editing and classifying them
 - Presenting data in tables
 - drawing diagrams and graphs for them
 - Calculating averages and measures of dispersions.

Remark: Descriptive statistics doesn't go beyond describing the data themselves.

Classification of Statistics ...

- **Descriptive Statistics (Example)**

- The average age of students in this class is 21.
- The sample shows 40% of year I students have positive attitude toward the delivery of lectures.
- Drawing graphs that show the difference in the 'scores' of fourth year Maths males and females students.

Classification of Statistics ...

- **Inferential Statistics**

- Deals with the method of inferring or drawing conclusion about the characteristics of the population based upon the results of a sample
- Utilizes sample data to make decision for entire data set based on sample
- Inferential Statistic (Example)
 - There is a definitive relationship between smoking and lung cancer
 - Drinking decaffeinated coffee can raise cholesterol levels by 7%.
 - Forward soccer players have a better performance than midfielders

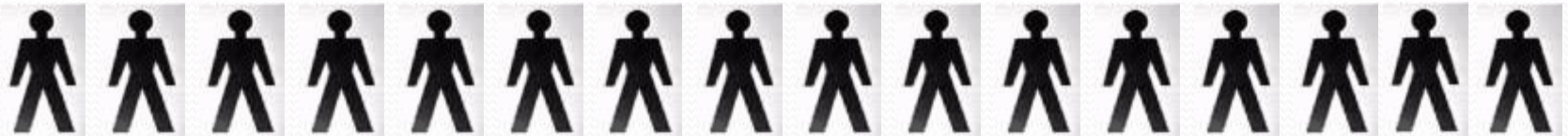
Definition of Some Basic Statistical Terms

- **Data**

- a collection of related facts and figures from which conclusions may be drawn
- a scientific term for facts, figures, information and measurement

- **Population/target population**

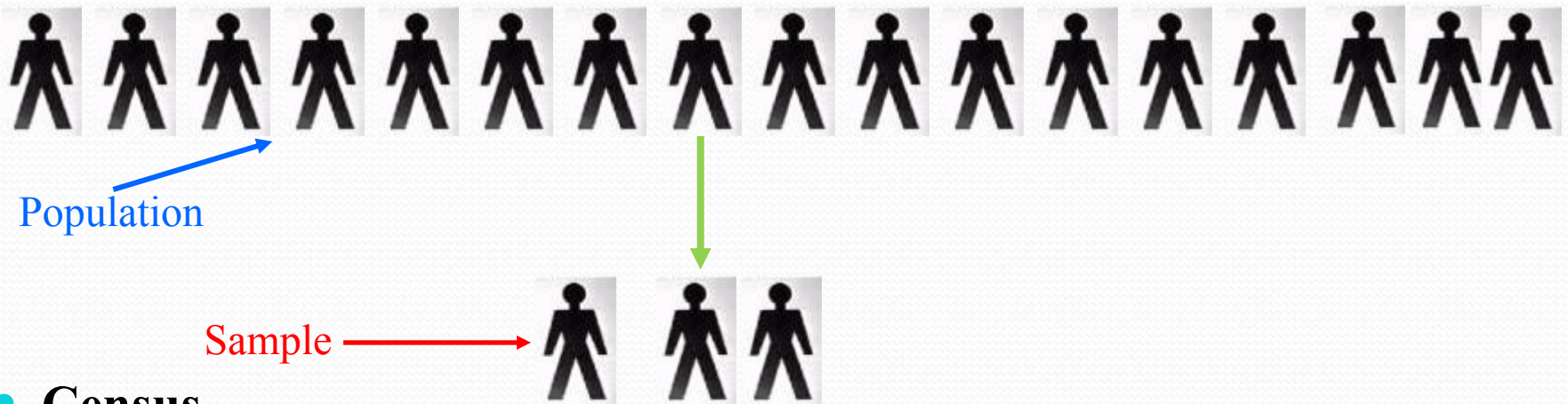
- a totality of things, objects, peoples, etc about which information is being collected
- Often too large to sample in its entirety
- Example: population of athletes fed a certain type of diet



Definition of Some Basic Statistical Terms

• Sample

- part of a population selected to draw conclusions about the population
- Subset of a population



• Census

- a complete enumeration of the population. But in most real problems it cannot be realized, hence we take sample.

Definition of Some Basic Statistical Terms

- **Statistic**

- A value computed from the sample, used to describe the sample.

- **Parameter**

- A descriptive measure (value) computed from the population.

- **Variable**

- is a characteristic or attribute that can assume different values.

- **Sampling frame**

- A list of people, items or units from which the sample is taken.

Stages in Statistical Investigation

- Statistical data must possess the following properties
 - The data must be aggregate of facts
 - They must be affected to a marked extent by a multiplicity of causes
 - They must be estimated according to reasonable standards of accuracy
 - The data must be collected in a systematic manner for predefined purpose
 - The data should be placed in relation to each other

Stages in Statistical Investigation

1. Data Collection

- The processes of measuring, assembling and gathering data
- Data may be collected by the investigator directly using interview, questionnaire, and observation or may be available from published or unpublished sources.
- Data gathering is the basis (foundation) of any statistical work.
- Valid conclusions can only result from properly collected data.

Stages in Statistical Investigation ...

2. Data Organization

- It is a stage where we edit our data
- The collected data involve irrelevant figures, incorrect facts, omission and mistakes
- classify (arrange) according to their common characteristics

3. Data Presentation

- The organized data can now be presented in the form of tables, diagram and graphs.
- The main purpose of data presentation is to facilitate statistical analysis

Stages in Statistical Investigation ...

4. Data Analysis

- Study the data to draw conclusions about the population parameter
- Dig out information useful for decision making
- Calculations of averages, the computation of measures of dispersion, regression and correlation analysis

5. Data Interpretation

- Draw valid conclusions from the results obtained through data analysis
- Making inference about general population from sample results

Uses and Limitations of Statistics

• Uses of Statistics

- Condenses and summarizes complex data
- Facilitates comparison of data
- Helps to measure variability in data
- Used to create relationship between variables
- Helps in predicting future trends
- Influences the policies of government
- Helpful in formulating and testing hypothesis and to develop new theories

Uses and Limitations of Statistics ...

- **Limitations of Statistics**

- Statistics doesn't deal with single (individual) values rather it deals with aggregate values
- Statistics can't deal with qualitative characteristics
- Statistical conclusions are not universally true
- Statistical interpretations require a high degree of skill and understanding of the subject
- Statistics can be misused

Scales of Measurement

- A variable in statistics is any characteristic, which can take on different values for different elements when data are collected
- Variable can be qualitative or quantitative
- **Qualitative Variables** are nonnumeric variables and can't be measured, example (gender, blood type, etc.).
- **Quantitative variables** are numeric variables and can be quantified
- Quantitative variables can be discrete (takes always whole number values) or continuous (assume or take any decimal value)

Scales of Measurement

- Measurement “is assigning numbers to objects, events, or abstract concepts according to a known set of rules”
- This permits data to be categorised, quantified and/or analysed in order that meaningful conclusions can be drawn.
- Four scales of measurement are identified

- Nominal Scale

Lowest Level

- Ordinal Scale

- Interval Scale

- Ratio Scale

Highest Level

Scales of Measurement

- **Nominal Scales of Measurement**

- A measure of identity or category into mutually exclusive classes
- Useful for quantifying qualitative data
- Provides no information regarding either order or magnitude
 - Example: Blood type (A, B, AB and O) , Name of A student, Identification number

- **Ordinal Scales of Measurement**

- A measure of order or rank
- Used to arrange data into series
- Provides no information regarding magnitude
 - Example: Ratings (good, v.good & excellent), economic status (low, medium & high)

Scales of Measurement ...

- **Interval Scales of Measurement**

- A measure of order *and* quantity
- Difference between values can be calculated
- Cannot establish 'x-fold' increase
 - Example: Temperature (10°C (50°F) and 20°C (68°F) as between 25°C (77°F) and 35°C (95°F))

- **Ratio Scales of Measurement**

- Highest level of measurement
- An interval scale with an absolute zero point
 - Example: weight, height, income, etc.

1.2. Methods of Data Collection and Presentation

Sources of Data

- **Primary data**

- data measured or collect by the investigator or the user directly from the source
- the data you collect is unique to you and your research and, until you publish, no one else has access to it
- The **primary sources** of data are objects or persons from which we collect the figures used for first hand information.

- **Secondary data**

- second-hand information and data or information that was either gathered by someone else
- The **secondary sources** are either published or unpublished materials or records.
- Few of sources of secondary data are

Sources of Data

- *government document, official statistics, technical report, scholarly journals, trade journals, review articles, reference books, research institutes, universities, hospitals, libraries, library search engines, computerized data base and world wide web (WWW).*
- **Before use of secondary data investigator should examine**
 - The type and objective of the situations.
 - The purpose for which the data are collected and compatible with the present problem.
 - The nature and classification of data is appropriate to our problem.
 - There are no biases and misreporting in the published data.
 - Reliability, homogeneity, and completeness.

Methods of Data Collection

- Planning to data collection requires
 - Identify source and elements of the data
 - Decide whether to consider sample or census
 - If sampling is preferred, decide on sample size, selection method, etc
 - Decide measurement procedure
 - Set up the necessary organizational structure
 - Collect data using different (appropriate) techniques

Methods of Data Collection...

1. Observation

- Recording the behavioral patterns of people, objects and events in a systematic manner.
- ranges from single visual observation to those requiring special skills like direct observation/examination.
- it may include laboratory experiment; conducting laboratory experiments on fields of chemical, biological sciences and so on.
- *Example:-* measuring height, weight, temperature, chemical component in water, performance of an athlete etc.

Methods of Data Collection...

2. Questionnaire

- It is a popular means of collecting data,
- it is difficult to design and often require many rewrites before an acceptable questionnaire is produced.
- a set of questions are administered (provided) to respondent either physically or through mail (Email, Postal, etc).
- Schedule through enumerations is the method in which investigator approach to the informant with prepared questionnaire and got replies to the questions.

Methods of Data Collection...

- **Advantages of Questionnaire**

- Can be used as a method in its own right or as a basis for interviewing or a telephone survey.
- Can be posted, e-mailed or faxed.
- Can cover a large number of people or organizations and wide geographic coverage.
- Relatively cheap and avoids embarrassment on the part of the respondent.
- Respondent can consider responses, and there is no interviewer bias.

Methods of Data Collection...

- **Disadvantage of Questionnaire**

- Historically low response rate (although inducements may help).
- Time delay whilst waiting for responses to be returned
- Several reminders may be required and it assumes no literacy problems.
- No control over who completes, and it is not possible to give assistance if required.
- Respondent can read all questions beforehand and then decide whether to complete or not.

Methods of Data Collection...

3. Interview

- A technique that is primarily used to gain an understanding of the underlying reasons and motivations for people's attitudes, preferences or behavior
- Interviews can be undertaken on a *personal (face to face)* or *via telephone (indirect method)*
- They can be conducted at work, at home, in the street or in a shopping center, or some other agreed location.

Methods of Data Collection...

• Advantages of Interviewing

- Serious approach by respondent resulting in accurate information and good response rate.
- Characteristics (motives and feelings) of respondent assessed – tone of voice, facial expression, hesitation, etc.
- Interviewer in control and can give help if there is a problem
- If one interviewer used, uniformity of approach.
- Completed and immediate.
- Possible in-depth questions.
- Can use recording equipment.
- Used to pilot than other methods.

Methods of Data Collection...

• **Disadvantages of Interviewing**

- Need to set up interviews.
- If many interviewers, training required.
- Time consuming.
- Geographic limitations.
- Can be expensive.
- Normally need a set of questions.
- Respondent bias – tendency to please or impress, create false personal image, or end interview quickly, Embarrassment possible if personal questions.
- Transcription and analysis can present problems (subjectivity).

Methods of Data Collection...

4. Extract from Records/Documentary Sources

- it is method of collecting information (secondary data) from published or unpublished sources
- Secondary data also collected from *diaries*
- **Advantage of secondary data**
 - Secondary data may help to clarify or redefine the definition of the problem as part of the exploratory research process.
 - Provides a larger database as compared to primary data
 - Time saving
 - Does not involve collection of data

Methods of Data Collection...

- **Disadvantages of secondary data**

- Lack of availability
- Lack of relevance
- Inaccurate data
- Insufficient data

5. Focus Group Discussion: discussion with selected individuals

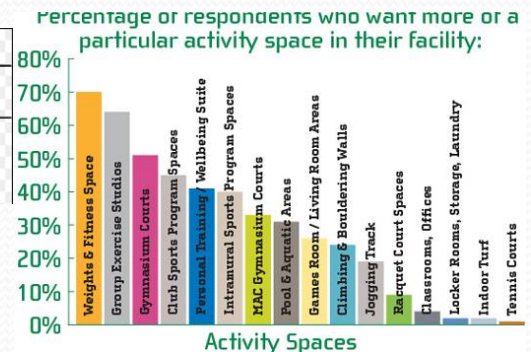
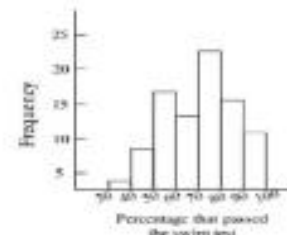
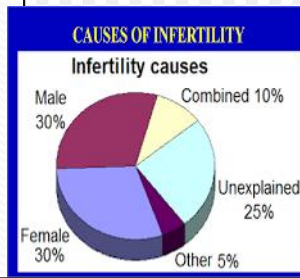


Methods of Data Presentation

- The major objectives of data presentation are
 - To presenting data in visual display and more understandable
 - To have great attraction about the data
 - To facilitate quick comparisons using measures of location and dispersion.
 - To enable the reader to determine the shape and nature of distribution to make statistical inference, and to facilitate further statistical analysis.
- There are three methods of data presentation

- Tables,
- Diagrams, and
- Graphs

Group	Body Weight(gm)	
	Initial	Final
Control	19.0 ± 1.5	26.6 ± 1.6
TPM	18.0 ± 2.1	27.0 ± 1.13



Methods of Data Presentation ...

- **Tabular presentation of data**

- Tables are important to summarize large volume of data in more understandable way.
- Tables can be
 - Simple (one way table): table which present one characteristics for example age distribution.
 - Two way table: it presents two characteristics in columns and rows for example age versus sex.
 - A higher order table: table which presents two or more characteristics in one table.

Methods of Data Presentation ...

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Methods of Data Presentation ...

- **Frequency Distribution**

- It is the organization of raw data in table form, using classes and frequencies.
- Frequency is the number of values in a specific class of the distribution.
- There are three basic types of frequency distributions
 - Categorical frequency distribution
 - Ungrouped frequency distribution
 - Grouped frequency distribution

Methods of Data Presentation ...

- **Categorical Frequency Distribution**

- *The categorical frequency distribution is used for data which can be placed in specific categories such as nominal or ordinal level data*
- *The major components of categorical frequency distribution are class, tally and frequency (or proportion).*
 - *Percentages are also usable*
- Forms of a categorical distribution

A	B	C	D
Class	Tally	Frequency	Percent

Methods of Data Presentation ...

- Example:** Data on smoking status by gender of a sample of 20 health workers in Jimma Hospital 1986 E.C was given. Construct categorical frequency distribution.

Observation	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Gender	M	F	M	M	F	F	F	M	M	M	F	F	F	F	M	F	M	F	M	M
Smoking status	Y	N	N	Y	N	N	Y	N	N	N	N	N	N	Y	Y	Y	N	N	Y	Y

Characteristics	Tally	Frequency
Gender		
Male		10
Female		10
Smoking status		
No		12
Yes		8

Methods of Data Presentation ...

- **Ungrouped Frequency Distribution**

- It is the distribution that use individual data values along with their frequencies.
- often constructed for small set of data on discrete variable (when data are numerical), and when the range of the data is small.
- sometimes it is complicated to use ungrouped frequency distribution for large mass of data, as result we use grouped frequency distribution.
- *The major components of this type of frequency distributions are class, tally, frequency, and cumulative frequency (less than/more than).*

Methods of Data Presentation ...

Example: Age in year of 20 women who attended health education at Jimma Health center in 1986 are given as follows. Construct ungrouped frequency distribution

30	25	23	41	39	27	41	24	32	29	29	35	31	36	33	36	42
35	37	41														

Age(x_j)	23	24	25	27	29	30	31	32	33	35	36	37	39	41	42
Tally	/	/	/	/	//	/	/	/	/	//	//	/	/	///	/
Frequency(f)	1	1	1	1	2	1	1	1	1	2	2	1	1	3	1

Methods of Data Presentation ...

- **Grouped Frequency Distribution**

- It is a frequency distribution when several numbers are grouped in one class
- the data must be grouped in which each class has more than one unit in width.
- We use when the range of the data is large, and for data from continuous variable.
- Sometimes used for large volume of discrete data

Methods of Data Presentation ...

- **Guidelines for classes**

- There should be 5 to 20 classes. Determine using Sturge's rule

$$K = 1 + 3.32 \log n$$

- Classes should be continuous.
- Classes must be mutually exclusive.
- Classes should be exhaustive.
- Classes should have same width (except open ended classes)

$$W = \frac{\text{Range}}{\text{Number of classes}} = \frac{R}{K}$$

Methods of Data Presentation ...

- **Class limit (CL)**

- It separates one class from another.
- The limits could actually appear in the data
- have gaps between the upper limits of one class and the lower limit of the next class.

- **Class boundary(CB)**

- Separate one class in a grouped frequency distribution from the other.
- The boundary has one more decimal place than the raw data.
- There is no gap between the upper boundaries of one class and the lower boundaries of the succeeding class.

Methods of Data Presentation ...

- **Unit of measurement (U)**

- This is the possible difference between successive values. E.g. 1, 0.1, 0.01 ...

- **Class width (W)**

- The difference between the upper and lower boundaries of any consecutive class.
- The class width is also the difference between the lower limit or upper limits of two consecutive classes.

- **Class mark (Midpoint)**

- It is found by adding the lower and upper class limit (Boundaries) and divided the sum by two.

Methods of Data Presentation ...

- Steps to construct grouped frequency distribution
 - Find smallest (S) and largest (L) values in your data
 - Compute difference between L and S, R
 - Determine the number of class using Sturge's rule, round up!
 - determine class width, ratio of R and K, round up!
 - Take the smallest value as the first class lower class limit, and add class width to get consecutive lower class limits
 - To get upper class limit subtract unit of measurement from second class lower class limit, and add class width to get remaining upper class limits
 - Subtract half of unit of measurement from lower class limit to get class boundary, and add half of unit of measurement to upper class limit to get upper class boundary
 - Tally data
 - Find cumulative frequency

Methods of Data Presentation ...

Example: Age in year of 20 women who attended health education at Jimma Health center in 1986 are given as follows. Construct grouped frequency distribution

30	25	23	41	39	27	41	24	32	29	29	35	31	36	33	36	42
35	37	41														

$$n=20$$

$$k=1+3.322(\log 20) = 1+3.322(1.3010) = 5.196 \Rightarrow k=6$$

$$w= (42-23)/54$$

The grouped frequency table using Sturges formula

Class	23-26	27-30	31-34	35-38	39-42
Frequency (f)	3	4	3	5	5

Methods of Data Presentation ...

- **Diagrammatic and Graphic presentation of the data**

- One of the most effective and interesting alternative way in which a statistical data may be presented is through diagrams and graphs.
- There are several ways in which statistical data may be displayed pictorially such as different types of graphs and diagrams.
 - Pie chart
 - Bar chart
 - Histogram

Methods of Data Presentation ...

● Pie Chart

- Pie chart is a circular diagram and the area of the sector of a circle is used in pie chart.
- To construct a pie chart (sector diagram), draw a circle (measures 360°)
- The angles of each component are calculated by the formula

$$\text{Angle of sector} = \frac{\text{Component part}}{\text{Total}} \times 360^{\circ}$$

- These angles are made in the circle by mean of a protractor to show different components.
- The arrangement of the sectors is usually anti-clock wise.

Methods of Data Presentation ...

- **Pie Chart (Example)**

- The following table gives the details of quarterly sale of a Sport Wear company's profit (in millions of dollar) in four quarters of a year.

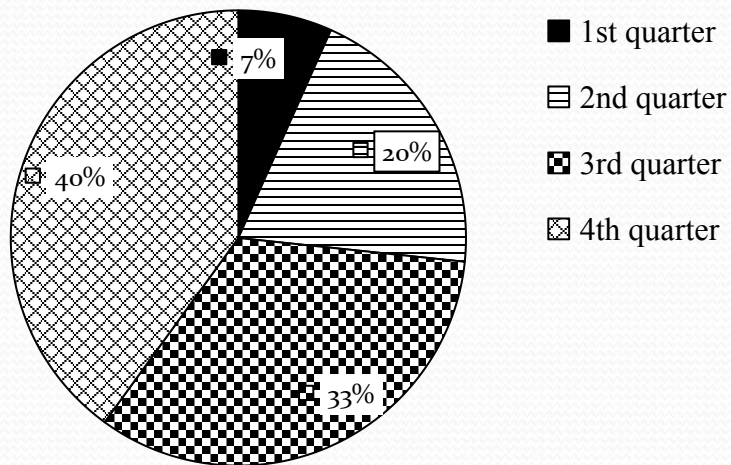
Month	Profit(\$,000,000)
1 st quarter	100
2 nd quarter	300
3 rd quarter	500
4 th quarter	600
Total	1500

- Construct a pie chart

Methods of Data Presentation ...

• Pie Chart (Example)

Quarter	Profit(\$,000,000)	Angle of sector (in degrees)	Percent (%)
1 st quarter	100	24	7
2 nd quarter	300	72	20
3 rd quarter	500	120	33
4 th quarter	600	144	40
Total	1500	360	100



Methods of Data Presentation ...

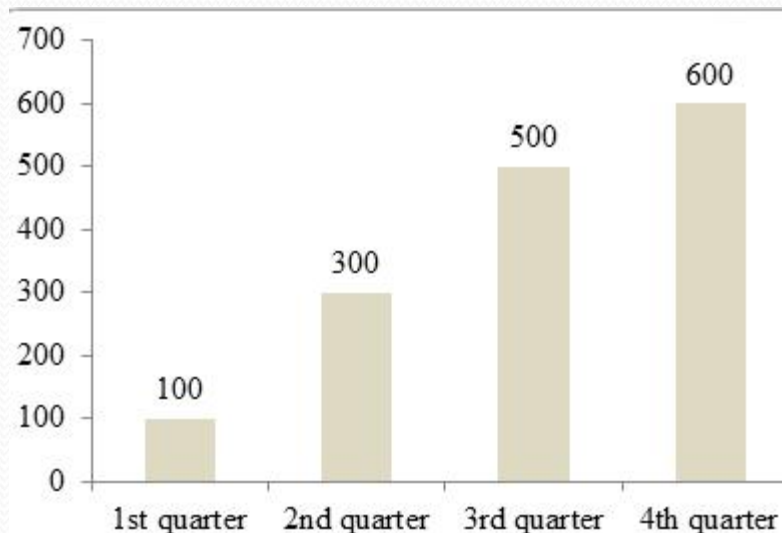
• Bar Chart

- Use vertical or horizontal bins to represent the frequencies of a distribution.
- While we draw bar chart, we have to consider the following two points.
 - Make the bars the same width
 - Make the units on the axis that are used for the frequency equal in size
- Bar charts can be
 - Simple bar chart,
 - Multiple bar charts,
 - Stratified or stacked bar chart
 - Deviation bar chart

Methods of Data Presentation ...

- **Simple Bar Chart**

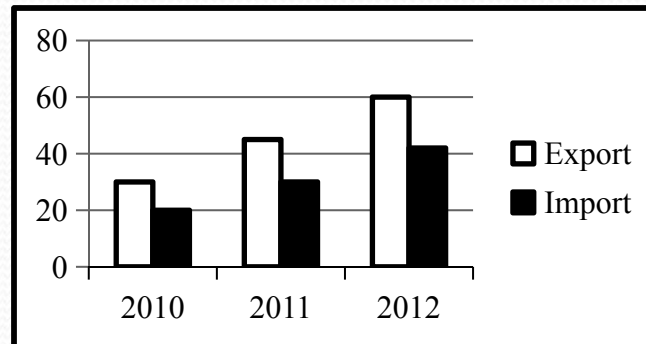
- Used to represents data involving only one variable classified on spatial, quantitative or temporal basis
- Make bars of equal width but variable length
- Example (Sports Wear company quarterly sales)



Methods of Data Presentation ...

• Multiple Bar Chart

- When two or more interrelated series of data are depicted by a bar diagram
- Make bars of equal width but variable length
- Example: Suppose we have export and import (in million) figures for a company working on mineral for few years.



Methods of Data Presentation ...

- **Stratified/Stacked Bar Chart**

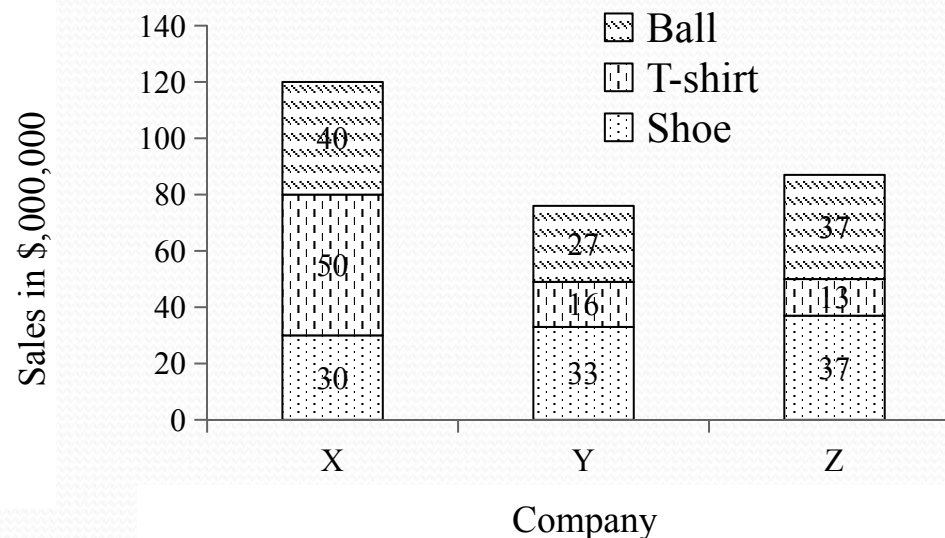
- *used to represent data in which the total magnitude is divided into different or components.*
- First make simple bars for each class taking total magnitude in that class and then divide these simple bars into parts in the ratio of various components
- Shows the variation in different components within each class as well as between different classes.
- Stratified bar diagram is also known as component bar chart.

Methods of Data Presentation ...

- **Stratified/Stacked Bar Chart**

- The table below shows the profit of a company (\$ Millions) from different item sales in 1st quarter of the year. Draw stratified/stacked bar chart

Company	Shoe	T-shirt	Ball	Total
X	30	50	40	120
Y	33	16	27	76
Z	37	13	37	87

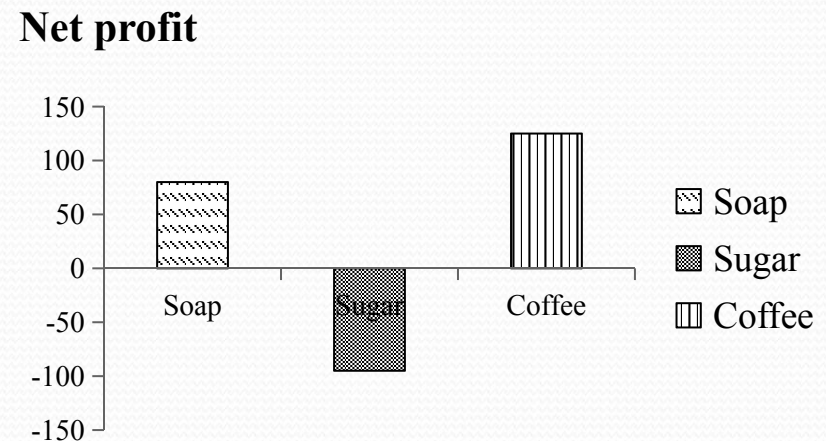


Methods of Data Presentation ...

• Deviation Bar Chart

- Used when the data contains both positive and negative values such as data on net profit, net expense, percent change etc
- Suppose we have the following data relating to net profit (percent) of commodity.

Commodity	Net profit
Soap	80
Sugar	-95
Coffee	125



Methods of Data Presentation ...

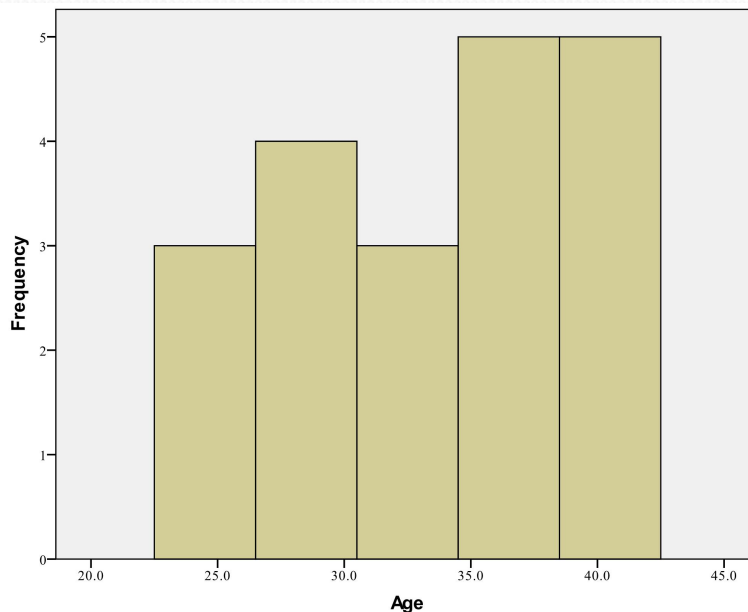
● Histogram

- *Histogram* is a special type of bar graph in which the horizontal scale represents classes of data values and the vertical scale represents frequencies.
- The height of the bars correspond to the frequency values, and the drawn adjacent to each other (without gaps).
- A graph which displays data by using vertical bars of various heights to represent frequencies.
- Class boundaries are placed along the horizontal axes.

Methods of Data Presentation ...

● Histogram

- A histogram shows the shape of continuous data, checks for homogeneity, and suggests possible outliers.
- To construct a histogram, we split the range of data into equal intervals, “bins,” and count how many observations fall into each bin.



Histogram for the age in years of 20 women